

What is Statistics?

Statistics is the branch of mathematics that deals with **collecting, organizing, analyzing, interpreting, and presenting data** to make informed decisions.

In **Data Science**, statistics is one of the most important foundations because it helps us:

- Understand data
- Find patterns and trends
- Make predictions
- Test hypotheses
- Build Machine Learning models
- Make data-driven decisions

SIMPLE DEFINITION

Statistics is the science of learning from data.

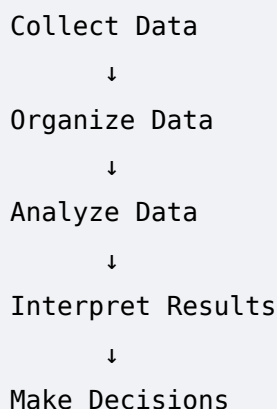
Real-Life Example

Suppose an online shopping company wants to know:

- Which products sell the most?
- What is the average customer age?
- Which city has the highest sales?
- Will a customer buy a product?

To answer these questions, the company uses **Statistics**.

Statistics Process



Types of Statistics

Statistics is divided into **two main types**.

1. DESCRIPTIVE STATISTICS

Descriptive Statistics is used to **summarize, organize, and describe** the data.

It only tells **what has happened**.

PURPOSE

- Summarize large data
- Understand data quickly
- Present data clearly

EXAMPLE

Student Marks

70
80
90
85
75

Instead of looking at every mark individually, we calculate:

- Average (Mean)
- Middle value (Median)
- Most repeated value (Mode)
- Highest value
- Lowest value
- Range
- Standard Deviation

These summarize the dataset.

TOPICS COVERED IN DESCRIPTIVE STATISTICS

A. MEASURES OF CENTRAL TENDENCY

These tell us the **center** of the data.

- Mean
- Median
- Mode

Example

Marks:

60 70 80 90 100

Mean

$$(60+70+80+90+100)/5 = 80$$

Median

80

Mode

No repeated value

B. MEASURES OF DISPERSION

These measure how spread out the data is.

Examples

- Range
- Variance
- Standard Deviation
- Interquartile Range (IQR)

Example

Student A:

50

51

49

50

50

Student B:

20

80

50

90

10

Both may have the same average, but Student B's marks vary much more.

C. FREQUENCY DISTRIBUTION

Shows how often values occur.

Example

Marks	Frequenc y
50	2
60	3
70	5

D. DATA VISUALIZATION

Descriptive statistics also includes charts.

Examples

- Bar Chart
- Pie Chart
- Histogram
- Box Plot
- Line Graph
- Scatter Plot

Applications of Descriptive Statistics

- Business reports
- Sales dashboards
- School result analysis
- Cricket statistics
- Weather reports
- Hospital reports

2. Inferential Statistics

Inferential Statistics is used to **draw conclusions or make predictions about a population using a sample.**

It tells us **what is likely to happen**.

POPULATION VS SAMPLE

POPULATION

Entire group.

Example

All Indian voters.

SAMPLE

Small portion selected from the population.

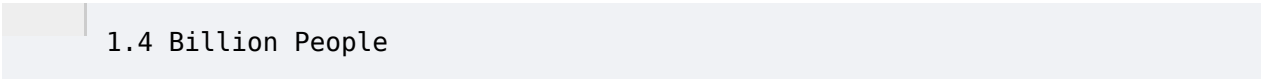
Example

1000 Indian voters.

We use the sample to estimate the opinion of the entire population.

EXAMPLE

Suppose India has

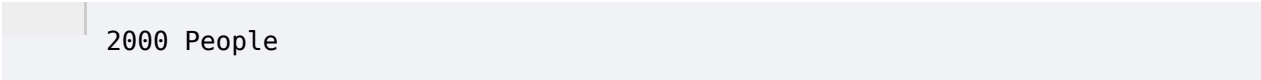


1.4 Billion People

Interviewing everyone is impossible.

Instead,

Survey



2000 People

Then estimate the opinion of the whole country.

This is Inferential Statistics.

TOPICS COVERED IN INFERENCE STATISTICS

A. SAMPLING

Selecting a subset from a population.

Types include:

- Random Sampling
 - Systematic Sampling
 - Stratified Sampling
 - Cluster Sampling
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B. ESTIMATION

Estimate unknown values.

Examples

- Population mean
 - Population proportion
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C. CONFIDENCE INTERVAL

Gives a likely range for the true population value.

Example

Average salary

₹45,000 ± ₹2,000

Meaning the true average is likely between ₹43,000 and ₹47,000.

D. HYPOTHESIS TESTING

Used to test assumptions.

Example

Company claims

"Our battery lasts 12 hours."

Statistics checks whether that claim is supported by data.

Examples of tests:

- Z-Test
 - T-Test
 - Chi-Square Test
 - ANOVA
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E. CORRELATION

Measures the relationship between variables.

Example

Hours Studied ↔ Marks

Positive correlation:

More study → Higher marks.

F. REGRESSION

Predicts one variable using another.

Example

Predict:

	House Price
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Using

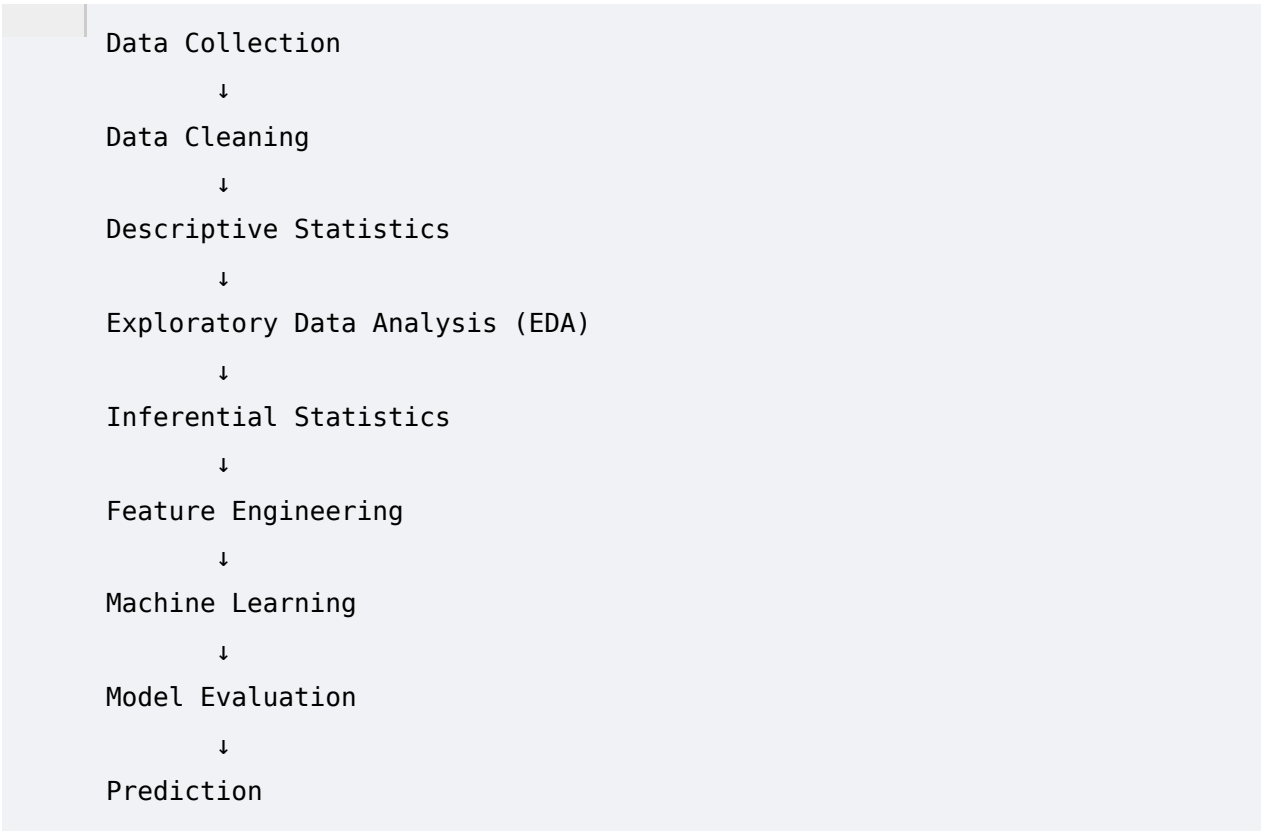
- Area
- Bedrooms
- Location

Difference Between Descriptive and Inferential Statistics

Feature	Descriptive Statistics	Inferential Statistics
Purpose	Describe data	Predict or infer from data
Data Used	Entire dataset	Sample data
Result	Summary	Conclusions about a population
Focus	What happened	What may happen
Examples	Mean, Median, Charts	Hypothesis Testing, Regression,

		Confidence Interval
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Statistics in the Data Science Workflow



Why Statistics is Important in Data Science

Statistics helps data scientists to:

- Understand datasets before modeling.
 - Detect missing values and outliers.
 - Summarize large amounts of data.
 - Identify trends and patterns.
 - Measure relationships between variables.
 - Test hypotheses with confidence.
 - Build and evaluate machine learning models.
 - Make reliable, data-driven business decisions.
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Common Statistical Concepts Every Data Scientist Should Learn

- 1. Mean
- 2. Median
- 3. Mode
- 4. Range
- 5. Variance
- 6. Standard Deviation
- 7. Probability
- 8. Sampling
- 9. Normal Distribution
- 10. Correlation
- 11. Covariance
- 12. Hypothesis Testing
- 13. Confidence Interval
- 14. Regression
- 15. Bayes' Theorem
- 16. Central Limit Theorem
- 17. p-value
- 18. Z-score
- 19. ANOVA
- 20. Chi-Square Test

SUMMARY

Descripti ve Statistics	Inferentia l Statistics
Summariz es and describes data	Draws conclusio ns and makes

	prediction s
Uses measures like mean, median, mode, variance, and charts	Uses sampling, confidenc e intervals, hypothesi s testing, correlation , and regression
Answers "What happene d?"	Answers "What is likely to happen?"
Commonl y used in Explorato ry Data Analysis (EDA)	Commonl y used for predictio n and decision- making

For **Data Science**, a good learning order is:

1. Introduction to Statistics
2. Types of Statistics
3. Population vs. Sample
4. Measures of Central Tendency
5. Measures of Dispersion
6. Probability
7. Probability Distributions

8. Sampling Techniques
9. Confidence Intervals
10. Hypothesis Testing
11. Correlation and Covariance
12. Regression Analysis
13. Statistical concepts used in Machine Learning